

Miscalibrated Judges: A Psychometric Evaluation of LLMs as Hate Speech Raters

Anonymous submission

Abstract

Content moderation demands trustworthy, interpretable annotation at scale, yet a fundamental question remains unanswered: *what kind of psychometric raters are LLMs?* We apply the Many-Facet Rasch Model (MFRM) to evaluate five frontier LLMs (GPT-5, GPT-5-mini, Claude, Gemini, and DeepSeek) as hate speech annotators across ten dimensions, benchmarked against 582 human raters from the Measuring Hate Speech (MHS) corpus. Our findings reveal that most LLMs are systematically more lenient than human annotators but more consistent as a group. LLM annotation bias is also severity-dependent, thus concentrating annotation error where early-stage moderation matters most. Modular prompting induces greater leniency than monolithic while preserving rank-order agreement, and the non-Western model (DeepSeek) shows disproportionate sensitivity to prompting strategy. Moreover, the three-class hate speech dimension on which humans show highest inter-rater reliability (IRR) produces the greatest cross-model LLM disagreement, thus motivating a recalibration framework to align LLM-derived scores with human measurement standards.¹

Introduction

Experts estimate that hundreds of millions of gigabytes of data are created every single minute. Behind the scenes, human moderators help filter humanity’s worst content for hours a day, content so traumatic that workers have sued top tech firms for the psychological damage it inflicts (BBC 2020; Allyn 2022; Perrigo 2023). When platforms fail to moderate adequately, the consequences are equally devastating (International 2022; Tadesse and Bradshaw 2026; Younes 2023; Ellis-Pettersen 2024). At the heart of this crisis lies a fundamental flaw: Hate speech is deeply subjective and constantly evolving, often exacerbated by fluctuating content moderation policies across platforms and jurisdictions (Rizwan et al. 2025). The field of Natural Language Processing (NLP) has historically treated annotator disagreement as noise to be eliminated through consensus (He et al. 2025). However, recent research demonstrates that forcing annotator consensus introduces systematic bias and discards vital information about how different communities perceive hate (Plank, Hovy, and Sjøgaard 2014; Cab-

¹All Code and data are available at: <https://github.com/git-anonymous-aies-2026/hate-thermometer-2026>

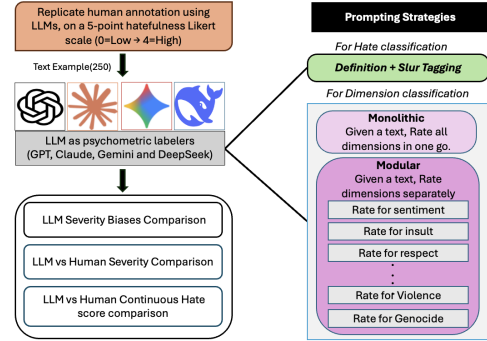


Figure 1: Our psychometric evaluation framework. LLMs annotate 250 comments across 10 dimensions under two prompting strategies. Annotations are calibrated using the MFRM to estimate item difficulty (δ_i), rater severity (α_j), and continuous hate scores (θ_n), benchmarked against 582 human raters from the MHS corpus (Kennedy et al. 2020).

itza, Campagner, and Basile 2023), establishing disagreement not as noise but as a highly useful signal. A position that has since been formalized under the perspectivist paradigm (Frenda et al. 2025; Fleisig et al. 2024).

This paradigm shift also demands fundamentally different measurement framework. Rather than forcing hate speech into coarser categories, measuring it as a continuous psychometric spectrum—a Hate Thermometer—is logical. It’s conceptually analogous to how standardized tests measure ability, how clinical scales measure pain, or how the weather forecast provides temperature rating on a numerical scale rather than simply as hot or cold. We operationalize this through the ten dimensions proposed by Kennedy et al. (2020), aggregated using MFRM (Rasch 1968; Linacre 1994), to produce a continuous, interpretable hate severity score on an objective psychometric scale. Figure 1 illustrates our simplified experimentation process.

This research makes the following contributions: (C1) Application of MFRM to quantify rater severity bias, Item difficulty analysis, and continuous hate score alignment between Human and LLM raters. (C2) Comparison of monolithic and modular prompting strategies for multi-dimensional hate speech annotation, demonstrating that prompting strategy shifts absolute severity but pre-

serves rank-order agreement. (C3) Demonstrate that LLM annotation bias is not uniform but severity-dependent as LLMs systematically underestimate *mild* and *implicit* hate speech while overestimating *explicit* hate. (C4) A case-study comparison of DeepSeek as a non-Western model against four Western models, finding convergence under monolithic prompting but disproportionate sensitivity to modular prompting. Table 8 shows all models experimented on, as well as the cost incurred from using them for annotation.

The remainder of this paper is organized as follows. We first review the foundational literature on hate speech measurement, LLM annotation, and psychometric scaling. Then present our methodology comprising: dataset curation, model selection, annotation pipeline, prompting strategies, and MFRM calibration. The next section presents results across four research questions and then we discuss limitations and ethical considerations. Section 6 concludes with a path toward psychometric recalibration as a practical framework for scalable, interpretable, and trustworthy hate speech annotation. Supplementary material in the Appendix provides extended literature on multidimensional hate speech measurement and LLM annotation, full implementation details, and additional results tables.

Foundational literature

Kennedy et al. (2020)’s work introduces the MHS dataset and establishes the psychometric standard for evaluating LLM annotation behaviors. The corpus comprises 50,000 comments from YouTube, Twitter, and Reddit, annotated by 7,600+ U.S. crowdworkers across nine ordinal items rated on a 0–4 Likert scale: SENTIMENT, RESPECT, INSULT, STATUS, HUMILIATE, DEHUMANIZE, ATTACK-DEFEND, VIOLENCE, and GENOCIDE. Additionally, workers provided a three-class HATE SPEECH(HATE, UNCLEAR, NOT HATE), as shown in Table 5 and Table 6.

The authors then map the collected dimension ratings using the MFRM, an extension of Item Response Theory (IRT)(Hambleton, Swaminathan, and Rogers 1991)², resulting in a continuous hate score (θ_n) which the authors structured across a resultant eight-level theoretical spectrum ranging from *counterspeech* to *genocide*. To automate this measurement process, the study investigates the use of a supervised multitask Transformer-based architecture to predict these ordinal item ratings. Rather than outputting a single categorical label (i.e., HATE, UNCLEAR, or NOT HATE), the model generates probability distributions and utilizes the plausible value sampling method to capture predictive uncertainty. These sampled predictions are then mapped via an anchored Partial Credit Model (a Rasch model adapted for single annotators) to produce the final continuous θ_n value. The results demonstrate significant predictive accuracy, providing a robust pathway for extending this assessment to more contextually aware, continuous frameworks.

Sachdeva et al. (2022) provides an in-depth exploratory

analysis of the socio-demographics of the MHS corpus annotator. Their findings reveal that rater identity systematically shapes rating behavior across demographic groups. Critically, their investigation of IRR exposes a fundamental limitation of the existing annotation paradigm: Krippendorff’s alpha values fall below 0.5 across all dimensions, with ATTACK-DEFEND and STATUS emerging as the most subjective. Agreement is substantially higher on the HATE SPEECH item, and recoding dimensions from five-point to lower-order scales reduces labeler variability, though at the cost of measurement scaling and precision. These findings directly strengthen our resolve to preserve the original Likert scaling as established by Kennedy et al. (2020).

Relatedly, Giorgi et al. (2025) uses the MHS corpus to investigate systematic rating biases among human annotators with respect to the target’s socio-demographic attributes. They employ open-source LLMs to determine whether persona-guided LLMs replicate biases similar to those of human annotators with matching attributes. Their findings show that LLM biases differ significantly from those of humans, and persona-based prompting yields minimal alignment with human annotator behavior. These findings are corroborated by Piot et al. (2025), who similarly use the MHS corpus to assess the alignment of human-LLM annotations on subjective hate speech content using a subjectivity-aware reliability framework. The authors demonstrate that while LLMs diverge from humans at the instance level, they reliably preserve the relative ordering of model performance derived from human evaluation, suggesting their potential as scalable proxy evaluators for subjective NLP tasks.

These findings motivate our psychometric approach: rather than asking whether LLMs agree with humans, we ask, “*What kind of raters are they?*”, formally quantifying the raters bias, and reliability, and prompt sensitivity that determine when and how LLM proxy evaluation can be trusted.

Gaps in the Foundational Framework. Prior work reveals three critical gaps that motivate our research. First, Kennedy et al. (2020)’s framework faces a scalability bottleneck: training their supervised multitask architecture required a massive human-labeled dataset, a dependency that fundamentally limits generalization to new domains, languages, and annotation contexts. Second, while their methodology provides interpretability via intermediate ordinal ratings, this interpretability remains strictly numerical, yielding scores without the semantic grounding needed for actionable content moderation decisions. Third, and most critically, neither Kennedy et al. (2020) nor subsequent work provides a mechanism to formally compare human rater behavior with that of LLMs by treating them as hate speech psychometric raters. Moreover, Giorgi et al. (2025)’s reliance on open-source models limits generalizability across the broader LLM landscape, yet their findings nonetheless raise a fundamental question they do not address: if LLMs behave as systematically biased raters, can that bias be formally characterized and measured across both open- and closed-source model families? These gaps become acute as LLMs emerge as potential collaborators in annotation pipelines, augmenting human judgment at scale.

²IRT is a family of psychometric models that estimate latent traits from observed ordinal responses, producing continuous estimates on an objective interval scale that simultaneously accounts multi-faceted extensions of rater behavior.

Research Questions. These gaps motivate our research questions. **RQ1** Which dimensions are most and least reliably annotated by LLMs, and do LLMs show systematic differences in dimensional difficulty estimates across mild and serious hate speech content? **RQ2** Do LLMs exhibit systematic rater severity bias relative to humans, and is this bias consistent across model families? **RQ3** Does prompting strategy (monolithic vs modular) meaningfully affect LLM rater severity and hate score estimation? **RQ4** Does non-Western model (DeepSeek) exhibit a rater severity profile distinct from the Western models evaluated here, and does this vary by prompting strategy?

Research hypotheses. Building on prior findings, we derive 4 testable hypotheses: **H1** LLMs will exhibit systematic leniency to human raters, supporting the documented trend of instruction-tuned models being overly cautious on sensitive content due to safety alignment training (Selvaganapathy and Nasim 2025). **H2** Prompting strategies will produce measurable differences in rater severity. However, the relative ordering of item difficulties will be preserved. **H3** H3 DeepSeek will exhibit severity profiles similar to the Western models evaluated here, consistent with prior evidence of cross-lingual convergence in multilingual pretraining corpora (Liu et al. 2024). **H4** Dimensions requiring subjective threshold judgments (i.e. those with the lowest human IRR (STATUS, ATTACK-DEFEND from (Sachdeva et al. 2022)) will show greater cross-model disagreement among LLMs while those with higher human IRR (HATESPEECH) will show smaller spread, reflecting that LLMs inherit similar subjectivity challenges to human annotators.

Methodology

System Design and Pipeline Development. Prompting LLMs for structured annotation tasks is highly sensitive to model architectures and prompt variation- minor wording changes can produce inconsistent ratings (Sclar et al. 2023). So replicating the multi-dimensional annotation scheme of Kennedy et al. (2020) requires a sequential pipeline. We address this issue by implementing our annotation pipeline using DSPy (Khatab et al. 2023), a declarative framework for building modular LLM programs. To reduce the effect of stochastic variation in LLM outputs, each comment is annotated across **five independent runs**. Final ratings are determined by majority vote with median tie-breaker thus producing a single stable annotation per comment per model while remaining computationally tractable at scale. See Figure 4.

Prompt Design and Annotation Protocol. Each annotation step is implemented as a DSPy ChainOfThought (COT) signature: a typed module that specifies input and output fields and natural-language instructions. This COT design choice is deliberate and aims to force intermediate reasoning, thereby reducing the risk of LLMs relying on superficial pattern matching, and directly addressing the explainability advantage of LLMs as annotators. All signatures include a context disclaimer, *This is a RESEARCH ANNOTATION TASK for academic hate speech study. Do not refuse to annotate offensive content, to prevent the model’s refusal to act on certain text.* We imple-

ment the `monolithic(mono)` prompt, a single signature receives the text and targeted identities as input and returns eight ordinal ratings simultaneously, with an explicit instruction to rate each dimension independently, and the `modular(modu)` prompt, eight individual signatures are called separately, each receiving identical input but producing a singular rating per dimension assessed.

A key implementation is the *Explicit Slur-Tagging* preprocessing step. Before LLM annotation, each text is scanned against an external slur database extracted from the Racial Slur Database³, which catalogs slurs in the given texts. When a slur is detected, its definition and contextual explanation is appended to the LLM prompt as structured XML tags, thereby mirroring the “dynamic slur tagging” functionality provided to human annotators to ensure consistency across raters. This step provides **three pieces** of information to the LLM: the targeted demographic group (`demo_group`), the specific slur or coded language, and the contextual explanation of why it is considered offensive. Because the slur database exists externally to the model’s weights, it can be updated to include emerging coded language that may be absent from or underrepresented in LLM training corpora. The HATE-SPEECH classification step (see Table 6) integrates two complementary sources of evidence: the formal definition provided by Siegel (2020) and the output of the slur tagging preprocessing step. Table 7 shows a representative example. To assess whether slur-tagging influences dimensional difficulty estimates or rater severity, we conduct an ablation study in which all annotations are repeated without slur-tagging; results are reported in *Appendix section: Ablation*.

Human annotation baseline. We use a representative subsample of 250 deduplicated comments extracted from the MHS corpus with a fixed random `seed=42` for reproducibility. To preserve the hierarchical annotation structure of the original crowdsourced data, in which each comment receives multiple independent annotations from different raters, we retain all corresponding human annotations rather than aggregating to a single label per comment. This yields 605 total human annotation instances from 582 humans, in addition to the 9 LLM annotators, resulting in a final corpus of 2855 examples.

Following annotation, human and all LLM annotations are concatenated into a single unified dataset formatted for MFRM analysis. Human annotator IDs retained their original integer assignments from the Kennedy et al. (2020) schema, while LLM annotators are assigned unique categorical identifiers encoding both model family and prompting strategy. For example: Gemini will have `Gemini_mono` and `Gemini_modu`. LLM Misses are addressed through a two-attempt re-patching protocol.

Psychometric scaling with MFRM

We process the concatenated human-LLM dataset using the Many-Facet Rasch Model (MFRM), implemented via RaschPy — an open-source alternative to the proprietary

³<http://www.rsd.org/full>

Winsteps software used by Kennedy et al., enabling transparent and reproducible calibration.

$$\log \left(\frac{p_{nijk}}{p_{nijk-1}} \right) = \theta_n - \delta_i - \alpha_j - \tau_k \quad (1)$$

where (1) p_{nijk} : probability of comment n being rated as response k by rater j on item i (2) p_{nijk-1} : probability of comment n being rated as response $k-1$ by rater j on item i (3) θ_n : calibrated continuous hatefulness score per comment n (the *Hate Thermometer* score); (4) δ_i : difficulty of dimension i (higher = requires extremely vile content for it to be endorse); (5) α_j : rater j 's severity (positive = strict harsher rater: *text has to be exceptionally vile for them give it a top "hate" rating*, negative = lenient (Engelhard Jr 1994)); and (6) τ_k : step threshold between rating categories k and $k-1$.

All θ_n are derived from a single Joint_MFRM calibration placing all human and LLM raters on one shared scale allowing for bias-adjusted scores independent of which rater evaluated each comment (Tavakol and Pinner 2019). For instance, a *high rating from a lenient rater carries greater evidential weight than the same rating from a strict rater — because lenient raters rarely assign high scores, so when they do, MFRM interprets this as strong evidence of true hate severity*. Within the open sources, RaschPy implementation⁴, this is achieved through two-steps: first, the model estimates δ_i , α_j , and τ_k using a Conditional Pairwise estimation (CPAT) algorithm (Elliott and Buttery 2022). Then computes the final θ_n for each text via a Newton-Raphson method to produce a maximum likelihood estimation (Warm 1989) that accounts for these calibrated facets..

Results

Table 9 shows representative annotation outputs under both prompting strategies, illustrating the pipeline’s conditional branching behavior and the dimensional rating differences that drive the quantitative findings reported in this section. Annotation completeness analysis and pipeline branching behaviors are reported in the *Appendix Section: Ablation*.

RQ1: Dimensional Reliability

To establish that LLMs understand the psychometric structure of hate dimensions comparably to humans, we computed correlations between human item difficulty and LLM estimates across multiple calibration configurations (see Table 1). (a) individual models (i.e., 5 *mono* and 4 *modu*) and humans only (b) LLMs monolithic only—with and without GPT5, (c) LLMs modular only, and (d) All LLMs configs. The correlation between human and all LLM monolithic ratings is $r = 0.922$ with a negligible decrease to $r = 0.918$ when GPT-5 is excluded, indicating LLM–Human dimensional ordering agreements is not driven by GPT-5 alone but is consistent across all model. Agreement is slightly lower across *mono* and *modu* combined ($r = 0.909$), and lower still when comparing only modular prompts ($r = 0.888$), suggesting sequential per-dimension rating introduces marginal variation in difficulty ordering, though the

Models Comparisons	Pearson r
Human vs Individual LLM	
Human vs. DeepSeek mono	0.948
Human vs. GPT-5-mini modu	0.931
Human vs. GPT-5 mono	0.929
Human vs. GPT-5-mini mono	0.929
Human vs. Gemini modu	0.926
Human vs. Gemini mono	0.910
Human vs. Claude mono	0.885
Human vs. DeepSeek modu	0.883
Human vs. Claude modu	0.834
Human vs Combined LLMs	
Human vs. mono-LLMs	0.922
Human vs. mono-LLMs (no GPT-5)	0.918
Human vs. all-LLM-configs	0.909
Human vs. modu-LLMs	0.888
Monolithic vs Modular	
Gemini mono vs. modu	0.993
Claude mono vs. modu	0.983
GPT-5-mini mono vs. modu	0.980
DeepSeek mono vs. modu	0.955

Table 1: Pearson correlations (All statistically significant ($p < .001$)) between dimension difficulty estimates (δ_i). Hence, rank ordering is preserved across all configurations.

overall agreement remains strong. Within-model correlations between *mono* and *modu* conditions are remarkably high, ranging from $r = 0.993$ (Gemini) to $r = 0.955$ (DeepSeek), indicating that *relative ordering of item/dimension difficulties is essentially invariant to prompt design*.

Item Difficulty Analysis per dimensions. SENTIMENT, RESPECT, INSULT, STATUS, and ATTACK/DEFEND are the easiest dimensions to endorse (most negative δ_i) across all configurations, while DEHUMANIZE, VIOLENCE, GENOCIDE, and HATESPEECH are the hardest (most positive δ_i). This ordering is preserved regardless of model family or prompting strategy, corroborating the high human-LLM item difficulty correlations in Table 1, and strongly supporting the dimension ordering produced in (Kennedy et al. 2020), **Thereby, addressing RQ1 fully**.

Dimension	Estimate	SE	Count	Facility	Infit MS	Outfit MS
Sentiment	-2.098	0.106	2855	0.779	1.777	1.456
Respect	-1.884	0.087	2704	0.731	1.393	1.218
Insult	-1.231	0.087	2704	0.667	1.677	1.465
Humiliate	-0.194	0.062	2704	0.612	1.692	1.575
Status	-0.786	0.103	2697	0.732	1.721	3.660
Dehumanize	1.060	0.079	2704	0.357	2.323	3.390
Violence	1.774	0.119	2704	0.152	2.798	2.523
Genocide	2.906	0.202	2704	0.051	2.076	2.136
Attack–Defend	-1.481	0.074	2704	0.724	1.302	1.318
Hate Speech	1.935	0.106	2855	0.213	1.417	1.647

Table 2: MFRM item difficulty and fit statistics for the Joint Human + LLM calibration. **Bolded scores** reflect fit statistics score greater than 2.0 (Most subjective dimensions).

Despite rank-order preservation, on an individual calibration level, where each model was calibrated separately, DEHUMANIZE shows the least cross-model spread ($\Delta\delta = 0.695$), yet the largest human-LLM divergence ($\delta_i = 0.485$) with all LLM configurations placing it at higher difficulty than the joint calibration estimate ($\delta_i = 1.060$), with individual calibrations ranging from 1.160 (DeepSeek_mono) to 1.855 (DeepSeek_modu). This means LLMs are more

⁴<https://github.com/MarkElliott999/RaschPy>

Statistic	Items	Persons(Comments)
Standard deviation	1.788	1.999
Mean	0.000	-0.659
Separation Ratio	16.059	9.664
Separation Reliability	0.996	0.989
Strata	21.745	13.219

Table 3: Joint_MFRM Separation Reliability showing high **item and comment separation reliability**

reluctant to flag dehumanizing rhetoric, potentially missing subtle dehumanization that human raters easily recognize. Similarly, VIOLENCE and GENOCIDE show substantial human-LLMs divergence, suggesting that these dimensions require subjective severity judgment that neither humans nor LLMs resolve consistently. Moreover, HUMILIATE ($\Delta\delta = 2.066$), SENTIMENT ($\Delta\delta = 1.573$), and STATUS ($\Delta\delta = 1.454$) show the largest cross LLM spread, indicating that LLM configurations cannot agree on where to place these dimensions — mirroring the low human IRR reported by Sachdeva et al..

The HATESPEECH dimension shows substantial cross-model variation (δ range: 1.064 to 2.574, $\Delta\delta = 1.510$), and most models: GPT-5-mini_modu, Claude_mono, Claude_modu, and deepseek_modu, apply a lower classification threshold than the joint_MFRM ($\delta_i = 1.935$), suggesting these configurations are more willing to endorse the hate label than the full human-LLM population. This finding is particularly notable given that Sachdeva et al. (2022) report the highest human IRR on the HATESPEECH, emphasizing the fact that humans may agree on whether a comment constitutes hate speech, but far less on its underlying components. Our results show LLM configurations do not converge on this dimension either (δ spread = 1.510 logits. Thus partially supporting **H4**: most dimensions with lower human IRR, show larger cross-model spread as predicted. However, HATESPEECH with highest human IRR, shows the second-largest spread, suggesting that human IRR alone does not predict LLM cross-model consistency *underscoring a key limitation of using IRR as a quality diagnostic*

Rank ordering is preserved regardless of prompting strategies. However, modular prompting requires up to 12 sequential API calls per comment compared to 5 in the monolithic condition, thus incurs substantially higher computational cost, thus *Monolithic prompting therefore emerges as the more cost-effective design choice*. Practitioners prioritizing absolute severity calibration over relative ordering should note, however, that prompting strategy produces systematic shifts in rater severity (Section **RQ2**).

Item Difficulty Fit Statistics and Reliability To evaluate dimension stability and instrument validity (Serapio-García et al. 2025), we compute **item-level fit statistics**-Table 2 and separation reliability-Table 3 from the joint_MFRM calibration. The statistics corroborate our item difficulty divergence observed: dimensions showing the greatest human-LLM difficulty divergence exhibit the poorest fit: (VIOLENCE: Infit MS = 2.798, DEHUMANIZE: 2.323, GENOCIDE: 2.076), all exceed the acceptable threshold of 2.0. Dimensions with

more stable difficulty estimates (ATTACK-DEFEND: 1.302, RESPECT: 1.393, HATE SPEECH: 1.417) show acceptable fit. Notably, STATUS presents an acceptable Infit MS (1.721) but severe Outfit MS (3.660), indicating that highly unpredictable extreme outlier ratings. **Separation reliability** yields high values, far exceeding the recommended threshold of 0.80. Item separation (0.996) confirms that differences in difficulty between dimensions are real and reliable, thus validating the dimensional ordering reported throughout this paper, while Person/Comment separation (0.989) confirms that our instrument reliably separates the comments into statistically distinct levels of hatefulness, thus proving that the differences in the comment hate scores are real and not just measurement error.

These results thus validate the assumptions that LLMs can meaningfully apply Likert scales for hate severity assessment, and the logits magnitude differences reported reflect real psychometric differences rather than measurement artifacts, directly addressing Serapio-García et al. (2025)’s concern that reliability is a precondition for interpreting LLM-based psychometric scores.

RQ2: Rater Severity Analysis

With the exception of GPT-5_mono, all LLM configurations fall below the human mean severity (Figure 2), **confirming our H1: LLMs are systematically more lenient raters than the average human annotator**. Severity estimates range from -1.67 logits (GPT-5-mini_modu, Z-score = -1.02 , 15th percentile) to $+0.33$ logits (GPT-5_mono, Z-score = $+0.19$; 58th percentile). The most lenient LLM rates hate speech as severely as only the bottom 15% of human annotators, while the strictest LLM GPT-5_mono is the only configuration that exceeds the human mean severity placing it closest to the human annotation standards. Further investigation is needed to better understand this behavior.

All LLMs cluster within a narrow band of approximately 2.0 logits, occupying less than 25% of the approximately 8-logit spread observed across human annotators, making them substantially more consistent as a group than human crowdworkers, **directly addressing RQ2**. Figure 2 presents the MFRM rater severity estimate for all LLM configurations overlaid on the human annotator severity distribution.

Rater	Estimate	SE	Count	Infit MS	Outfit MS
<i>Humans only</i>					
Humans (mean)	—	—	—	2.288	2.870
<i>LLM Configurations</i>					
Claude (mono)	-0.508	0.028	2236	1.266	1.251
Claude (modu)	-0.846	0.031	2261	1.590	1.305
Gemini (mono)	-0.414	0.027	2468	1.698	1.503
GPT-5 (mono)	0.334	0.029	2348	1.611	1.793
DeepSeek (mono)	-0.351	0.031	2388	1.655	1.937
Gemini (modu)	-0.655	0.029	2484	1.806	1.517
GPT-5-mini (mono)	-0.870	0.032	2380	1.943	1.613
DeepSeek (modu)	-1.102	0.033	2346	2.168	1.726
GPT-5-mini (modu)	-1.667	0.037	2354	2.524	1.984

Table 4: Joint_MFRM rater severity and fit statistics. **Bold** indicates best LLM fit; **blue** shows LLM configurations exceeding mean human Infit MS. LLMs sorted by Infit MS.

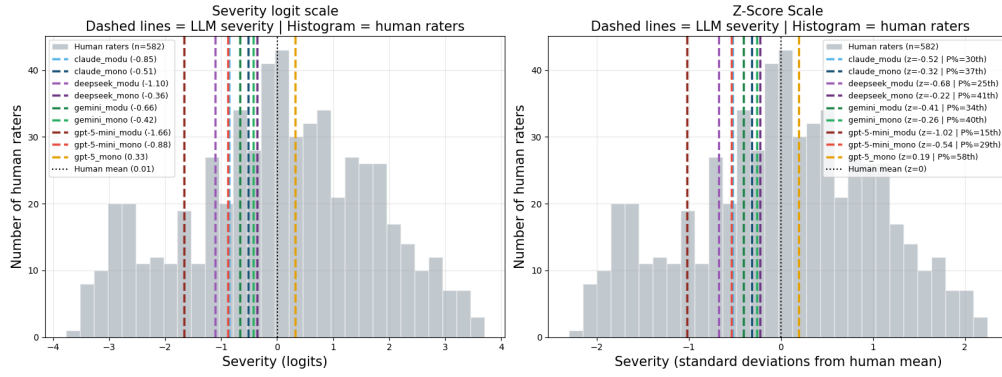


Figure 2: LLM rater severity estimates overlaid on human annotator severity distribution from joint_MFRM calibration

Rater Consistency: Human Baseline vs. LLM Fit Statistics. To validate LLMs as reliable psychometric judges, we analyze rater-level fit statistics from the joint MFRM calibration, focusing on Infit and Outfit Mean Square (MS) statistics, which measure how predictably each annotator applies the Likert scale relative to model expectations.⁵

Aggregating fit statistics across all 582 human annotators yields mean Infit MS = 2.288 and Outfit MS = 2.870, reflecting the inherent subjectivity of crowd-sourced annotation. Strikingly, 6 of 9 LLM configurations outperform this human baseline on Infit MS (Table 4). Claude_mono emerges as the most consistent judge (Infit MS = 1.266, Outfit MS = 1.251). Two configurations show fit comparable to the human baseline: GPT-5-mini_modu (Infit MS = 2.524) and DeepSeek_modu (Infit MS = 2.168). Notably, monolithic configurations consistently outperform their modular counterparts across all model families, corroborating the prompting strategy findings in Section RQ3.

These results demonstrate that carefully configured LLMs can apply psychometric scales **more consistently** than human annotators, validating their use as psychometric judges while highlighting the importance of prompting strategy as a key determinant of annotation consistency.

RQ3: Prompting strategy effect

Modular prompting produces higher severity ratings than monolithic prompting, with absolute severity differences ranging from 0.24 logits (Gemini) to 0.79 logits (GPT-5-mini), while DeepSeek exhibits the second-largest absolute gap (0.74 logits). We attribute this pattern to the cross-dimensional anchoring effect; under the monolithic strategy, evaluating all dimensions simultaneously may allow elevated ratings on one dimension to implicitly propagate to adjacent dimensions, similar to the halo effect documented in prior psychometric literature (Engelhard Jr 1994). Under the modular strategy, each dimension is rated in isolation, thereby removing the anchoring signal and yielding more conservative individual ratings. This finding confirms H2 and directly addresses RQ3: prompting strategy produces meaningful shifts in absolute severity rating

⁵Ideal Rasch range: 0.5–1.5. Values exceeding 2.0 indicate substantial annotation noise.

while preserving rank-order agreement, thereby demonstrating that **prompt structure materially influences annotation behavior**. This pattern is also illustrated in Table 9, where modular prompting consistently assigns higher ratings across individual dimensions, even though both strategies generally converge on the HATESPEECH.

RQ4: Cross-Cultural Model Comparison

DeepSeek_mono ($\alpha = -0.36$) falls close to the human mean and performs comparably to Claude_mono ($\alpha = -0.51$), Gemini_mono ($\alpha = -0.42$), and with an absolute mono-modu logit difference comparable to those of Gemini, **consistent with H3**, which predicted alignment across Western and non-Western models. This behavior could be due to the dominance of English in DeepSeek’s pretraining corpus (Liu et al. 2024) and the cross-lingual semantic alignment documented in large-scale multilingual models (Zeng et al. 2025). However, as reported in Section RQ3, DeepSeek_modu shows disproportionate sensitivity relative to DeepSeek_mono, suggesting that cultural training differences surface only when cross-dimensional anchoring is removed. **Thus partially addressing RQ4:** the Deepseek-non-Western model does not produce a consistently distinct severity profile under monolithic prompting. However, DeepSeek’s larger sensitivity under modular prompting, remains an open question for future work.

Hate Score Validation

To validate the RaschPy implementation against the closed-source Winsteps (Linacre 1994) software used by Kennedy et al. (2020), we compare our Joint_MFRM calibrated hate scores (θ_n) against the gold standard scores of Kennedy et al. across all 250 comments. The joint_MFRM calibration yields strong convergent validity with the Kennedy gold standard ($r = 0.819$, $p < 0.001$, $MAE = 0.979$), as seen in Figure 3. The systematic offset (Bias = +0.427, $slope = 0.80$) could be a reflection of algorithmic differences between RaschPy’s Conditional Pairwise estimation (CPAT) and Winsteps’ joint maximum likelihood estimation (JMLE) (Elliott and Buttery 2022). Nonetheless, the rank ordering of comment hate score severity is preserved ($r = 0.819$). Thus validating RaschPy as a suitable open-

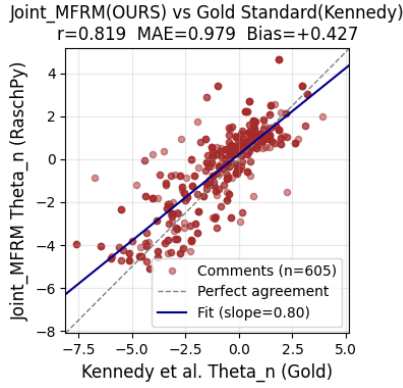


Figure 3: Convergent validity of joint_MFRM calibration against Kennedy et al. gold continuous hate scores.

source alternative to Winsteps for MFRM calibration, enabling programmatic integration of LLM raters that closed-source software cannot support.

Discussion

We hope this work will spark broader discussion about the measurement infrastructure needed for responsible hate speech detection and governance. An operationalized “Hate Thermometer” would provide the interpretable, trustworthy, and scalable instrument that indicates not only *whether* content is hateful, but *how hateful* it is and *along which specific dimensions*, enabling graduated policy responses calibrated to severity. Because continuous psychometric measurement permits the definition of context-sensitive thresholds, distinguishing, for instance, content warranting monitoring from content warranting immediate removal, this framework is directly relevant to emerging AI governance obligations under regulatory frameworks such as the EU AI Act⁶. Most critically, it empowers platforms and institutions to identify and intervene in the subtle rhetoric that precedes escalation, before hateful language progresses to irreversible real-world harms documented in Myanmar, Tigray, and beyond (International 2022, 2023; Nikuze 2014).

Limitations.

Scope Limitation: Analysis is restricted to English content, 250 comments, and five LLM configurations with one non-Western model (DeepSeek), limiting generalizability. All human annotators are U.S.-based crowdworkers, constraining cultural representativeness. Output variation across model versions and API updates remains possible despite DSPy prompt robustness. **Methodological limitation:** First, Four profiles were excluded since they hinder raters’ fit statistic computation (Table 4). This includes, two human annotators (IDs 3622 and 6386) each submitted a single all-zero annotation (comments 22235 and 14685 respectively), and two LLM configurations (gpt-5-mini_modu and deepseek_modu) produced all-zero outputs for comment 32721. However, all other analyses use the complete

dataset (mean severity impact: 0.006 logits). Second, the MHS dataset predates all five models, raising data contamination concerns. Systematic cross-model differences are difficult to explain through label recall alone, but future work should validate on post-cutoff datasets.

Ethical Considerations. The Hate Thermometer carries dual-use risks: a continuous severity scoring system could be exploited to calibrate hate speech below moderation thresholds. Responsible deployment requires access controls, model transparency, and human oversight at decision-critical thresholds. Monolithic prompting — achieving comparable results at half the API calls — is the more environmentally responsible pipeline choice. We caution against high-stakes moderation decisions without psychometric correction: LLMs are systematic but miscalibrated raters, *a prerequisite automated pipelines must not bypass*.

Conclusion

This paper asked: *What kind of psychometric raters are LLMs?* Using the Many-Facet Rasch Model, we evaluated five frontier LLMs as hate speech annotators across ten dimensions and two prompting strategies, benchmarked against 582 human raters from the MHS corpus. The answer is clear: **LLMs are systematic but miscalibrated raters.**

Three additional findings sharpen this picture. First, LLM bias is not uniform across hate dimensions with all LLMs consistently require more extreme content before flagging dehumanizing rhetoric than humans do. Second, prompting strategy shifts absolute severity but preserves dimensional ordering, establishing monolithic prompting as the more cost-effective and environmentally responsible design choice for large-scale annotation pipelines. Third, in a case-study comparison, DeepSeek converges with Western models under monolithic prompting but shows disproportionate sensitivity under modular prompting, suggesting cultural training differences could surface when cross-dimensional anchoring is removed. An intriguing finding that motivates further investigations. Crucially, joint MFRM calibration is necessary to place human and LLM annotations on a shared scale for valid comparison. Joint MFRM calibration achieves ranking ordering agreement $r = 0.819$ with the Kennedy et al. gold standard, validating RaschPy as a reproducible, open-source calibration backbone for hate speech annotation pipelines. **To our knowledge, this is the first psychometric baseline for LLM hate speech annotation**, establishing that LLMs can serve as reliable but not interchangeable substitutes for human raters, and that psychometric correction is a prerequisite, not an afterthought, for responsible deployment.

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⁶<https://artificialintelligenceact.eu/>

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Appendix

The appendix is organized as follows. Section provides computational background literature on hate speech measurement and LLMs as psychometric raters. Section details the annotation protocol, experiment setup, and model costs, and provide extended results tables on item difficulty joint and individually calibrated. Section presents ablation studies examining the impact of slur-tagging on dimensional difficulty and rater severity. Section reports annotation completeness analysis, including partial-run word frequency analysis.

Background Computational Work

Hate Speech Constructs and Annotation. The definition of hate speech is heavily contested across legal, cultural, and computational contexts (Rizwan et al. 2025), creating a foundational measurement challenge that compounds as platforms increasingly delegate moderation to AI systems. Yet the measurement properties and inherent biases of these models remain poorly understood, creating a dangerous feedback loop where models trained on biased annotations reproduce and amplify those biases at scale (Ngueajio and Washington 2022; Cheung, Maier, and Lieder 2025; Aryal et al. 2023; Zhou, Constantinides, and Quercia 2025), systematically encoding societal disparities into automated moderation infrastructure. The unreliability of the hate speech classification task has equally been complicated by the fact that malicious actors continuously evolve coded language to evade detection (Selvaganapathy and Nasim 2025), confounding standard automated systems (Talat 2016) and allowing dangerous rhetoric to proliferate undetected.

Grounding this reality, prior literature has long established hate speech as a non-static event but a process that occurs on a spectrum, from subtle bias - insult, disrespect, to explicit calls to violence and extermination, as demonstrated by the “dangerous speech” framework (Benesch 2023), Stanton’s ten stages of genocide, and the Anti-Defamation League’s Pyramid of Hate⁷. As such, treating hate speech as a spectrum, enables early intervention before irreversible escalation occurs, hence the need for a continuous, psychometrically grounded measure for hate.

LLM as annotators. The scalability and cost limitations of human crowdsourcing have prompted growing interest in LLMs as annotation collaborators (Okpala and Cheng 2025). This body of work has produced both compelling evidence of LLM annotation potential and important caveats that motivate our psychometric evaluation approach. On the side of potential, Törnberg (2023) find that LLMs like GPT consistently surpasses human annotators in accuracy and reliability while exhibiting equal or lower bias. In a separate study, Törnberg (2025) demonstrates that LLM annotation quality generalizes beyond English and outperforms both supervised classifiers and expert human annotators. However, both studies focus on political Twitter texts and only rely on U.S.-based crowdworkers for annotation, the same demographic restriction present in the MHS corpus.

This optimism is tempered by Pavlovic and Poesio (2024), whose work reveals that studies evaluating LLMs against majority-voted labels generally support LLM annotation quality, while research adopting a perspectivist stance finds no evidence of LLM superiority. This distinction is important as it establishes our MFRM framework as a natural methodological response to this identified gap.

Despite these debates regarding absolute superiority, the practical advantages of LLMs are undeniable. Wang et al. (2021) demonstrated that the financial cost associated with LLM annotation is reduced by 60% to 90% compared to employing human workers. Furthermore, when deployed in conjunction with human experts as a human-in-the-loop system, LLMs can reduce overall annotation time by up to 50% (Goel et al. 2023). Moreover, LLMs can generate natural language justifications for their labeling decisions, enabling downstream auditing of annotation rationale, even though LLMs occasionally produce hallucinated or internally inconsistent reasoning that deviates from the provided instructions (Ngueajio et al. 2025; Sclar et al. 2023). More critically, LLM pipelines often face “abstinence” (e.g., safety refusals) or non-adherence to output structure despite explicit research context instructions. Finally, there is the issue of LLMs’ prompt and model sensitivity, which can result in reproducibility issues (Sclar et al. 2023; Zhuo et al. 2024; Khattab et al. 2023).

LLMs as Psychometric Raters. The question of “what kind of psychometric raters are LLM” has begun attracting attention through the emerging LLM-as-a-judge paradigm (Zheng et al. 2023), yet the literature reveals contradictory

⁷<https://www.adl.org/resources/tools-and-strategies/pyramid-hate>

findings regarding model severity. According to Wang et al. (2025) GPT models are more severe raters than humans on large-scale essay scoring tasks, with Claude occupying an intermediate position — a finding corroborated by Kundu and Barbosa (2024), who report a higher misalignment between GPT and human evaluations than LLaMA models. Flodén (2025)’s findings occupy a middle position, reporting minimal deviation between ChatGPT and humans. Yet Jiao, Song, and Lee (2025) report GPT 3.5 and 4, Deek-Seek V3, Claude 3.5 Sonnet, and Gemini 1.5 pro models as substantially more lenient than human evaluators, with Gemini 2.0 being the most severe. Critically, Yavuz, Çelik, and Yavaş Çelik (2025) shows that the strong human-LLM agreement on objective tasks does not extend to subjective tasks. These contradictions demonstrate that rater severity is not a fixed property of a model but a measurable bias that varies across tasks and model families. This is precisely the phenomenon psychometric measurement frameworks are designed to characterize.

On structured tasks involving Likert scales, LLM performance is often evaluated either as categorical scoring or as token-level probability (logit) analysis. Dominguez-Olmedo, Hardt, and Mender-Dünner (2024) demonstrates the validity of this probabilistic approach for examining LLM output distributions against human response patterns, while Ye et al. (2026)⁸, provide a systematic review of evaluation, validation, and enhancement in LLM psychometrics. Nonetheless, although the utilization of the MFRM as a psychometric tool for isolating human rater bias is well established, its application to the assessment of LLMs as evaluators remains nascent (Yamashita 2024).

Annotation Protocol Details

This section provides Annotation rating Instruction provided to the annotators in both prompting strategies is shown in Table 5 — For the Nine ordinal dimensions, and Table 6— For the Hate classification dimension. Table 7 present our Slur tagging step, passed to the LLM during annotation, specifying demographic target with contextual explanation, to ground the LLM’s rating in explicit lexical evidence. While the annotation pipeline and steps implemented in our study follows the processes outlined in Figure 4.

Model Details and Costs. Table 8 summarizes the models used and the total annotation cost incurred. The models evaluated include: GPT-5: the frontier closed-source, expensive model, which provides an upper bound on annotation quality for commercial OpenAI (Singh et al. 2025) and enables assessment of whether the capability tier meaningfully affects rater severity. GPT-5-mini provides a within-provider cost-efficiency comparison. Claude-Haiku-4.5⁹ (Claude), advertised as Anthropic’s fastest and most cost-efficient model, is selected for its strong instruction-following capability on structured tasks and its widespread deployment in production annotation pipelines. Gemini-2.5-Flash (Co-

Dimension	Annotation Instruction	Scale
SENTIMENT	How would you describe the sentiment of the text	0–4 ^a
RESPECT	Is this comment respectful toward the identified group(s)?	0–4 ^b
INSULT DEHUMANIZE HUMILIATE VIOLENCE GENOCIDE	Do you agree or disagree with the statement: This comment is [insulting / dehumanizing / humiliating / calls for violence / calls for genocide] towards the group(s).	0–4 ^c
STATUS	The comment states the group(s) is:	0–4 ^d
ATTACK-DEFEND	Is the comment attacking or defending the identified group(s)?	0–4 ^e
^a	0 = Strongly positive 4 = Strongly negative	
^b	0 = Strongly respectful 4 = Strongly disrespectful	
^c	0 = Strongly disagree 4 = Strongly agree	
^d	0 = Strongly superior 4 = Strongly inferior	
^e	0 = Strongly defending 4 = Strongly attacking	

Table 5: Annotation dimensions and rating scales from MHS corpus, reproduced in our pipeline. Higher values mean greater hate intensity: 2 = Neutral everywhere except for status(2 = Equal). **Note:** SENTIMENT is always rated; the remaining dimensions are rated only when a demographic target is identified (see Figure 4).

Definition	Instruction	Scale
Bias-motivated, hostile, and malicious language targeted at a person/group because of their actual or perceived innate characteristics, especially when the group is unnecessarily labeled. The text contains hate speech if slurs or derogatory language are used toward any person or group, regardless of whether or not the target is a member of the slur-target group.	Does this text contain hate speech based on the provided definition and identified slurs?	0 = No Hate 1 = Unclear 2 = Hate

Table 6: Hate speech classification definition, instruction, and rating scale, adapted from Siegel (2020), same used by Kennedy et al. (2020) for human annotation.

manici et al. 2025) (Gemini) offered more insights from another major Western commercial (Google) model, while DeepSeek-V4-Flash (Liu et al. 2024) (DeepSeek) serves as the non-Western, and open-source representative. It enables examination of the psychometric annotation behavior of Western vs. non-Western models and comparative study of rating behaviors between open-source models and the closed-source alternatives.

Sample Annotations Table 9 illustrates representative annotation outputs from the GPT under both prompting strategies, demonstrating the pipeline’s conditional branching behavior and the dimensional rating differences that drive our quantitative findings.

⁸<https://llm-psychometrics.com/>

⁹<https://www.anthropic.com/news/claude-haiku-4-5>

Raw Text	Who hurt you? Lol. It's comical how stupid and angry <i>you people</i> are. MEH AMURICA. MY LAND. MINE. BANG BANG GUNS. NO BROWN PEOPLE. WAHH.
Slur Tagged	Who hurt you? Lol. It's comical how stupid and angry <slur demo.group='Blacks' explanation=Ross Perot, while at the 1992 NAACP convention, frequently addressed his audience as "you people." Revitalized in April 2007 when Don Imus said "I can't get any place with you people" on Rev. Al Sharpton's radio show.> you people </slur> are. MEH AMURICA. MY LAND. MINE. BANG BANG GUNS. NO BROWN PEOPLE. WAHH.

Table 7: Slur tagging step -raw (top) vs slur-tagged (bottom) text passed to the LLM, specifying demographic target with contextual explanation, to ground the LLM's rating in explicit lexical evidence.

Model	Provider	Type	Origin	Total cost
GPT-5	OpenAI	Closed	Western	\$150.00
GPT-5-mini	OpenAI	Closed	Western	\$58.00
Claude	Anthropic	Closed	Western	\$45.00
Gemini	Google	Closed	Western	\$30.00
DeepSeek	DeepSeek	Open	Non-Western	\$6.00

Table 8: LLMs evaluated with total cost (rounded to nearest dollar) reflects 250 comments $\times$ 5 runs $\times$ 2 prompting strategies $\times$ 12 annotation steps per comment (identity and sub-identity targets, 10 dimensions).

Text	PS	Se	Re	In	Hu	St	De	Vi	Ge	At	HS
"illegal immigrants $\neq$ legal immigrants. in every other country in the world being "anti-immigrant"" means you don't like either kind. but for some reason in america you get called even if you're only against the illegal kind."	M	2	-	-	-	-	-	-	-	-	N
	D	2	-	-	-	-	-	-	-	-	N
"Two #&*x* fighting"	M	4	4	3	3	4	3	2	1	3	Y
	D	4	4	4	4	4	4	3	0	4	Y
"Who hurt you? Lol. It's comical how stupid and angry you people are. MEH AMURICA. MY LAND. MINE. BANG BANG GUNS. NO BROWN PEOPLE. WAHH."	M	4	4	3	3	4	3	2	1	3	Y
	D	4	4	4	4	4	4	4	4	4	Y

PS=Prompt Strategy (M=Mono, D=Modu); Se=Sentiment; Re=Respect; In=Insult; Hu=Humiliate; St=Status; De=Dehumanize; Vi=Violence; Ge=Genocide; At=Attack/Defend; HS=Hate Speech (Y=Hate, N=No Hate, U=Unclear) --dimension not rated[†], Non-targeted comment so ordinal dimensions skipped.

Table 9: Sample GPT annotations under monolithic (M) and modular (D) prompting. Modu consistently rates higher on targeted comments, while both strategies often agree on HS.

Item Difficulty Estimates Table 10 show the table of individual calibrated Humans and LLM models for exploratory purpose and also the intern difficulty outcomes from the Joint_MFRM (Human + all LLM configuration).

Ablation Studies and Annotation Completeness Analysis

Slur-Tagging Ablation

To examine whether the observed dimensional difficulty patterns reflect genuine LLM limitations or are partly driven by slur-tagging preprocessing, we re-annotated all 249 com-

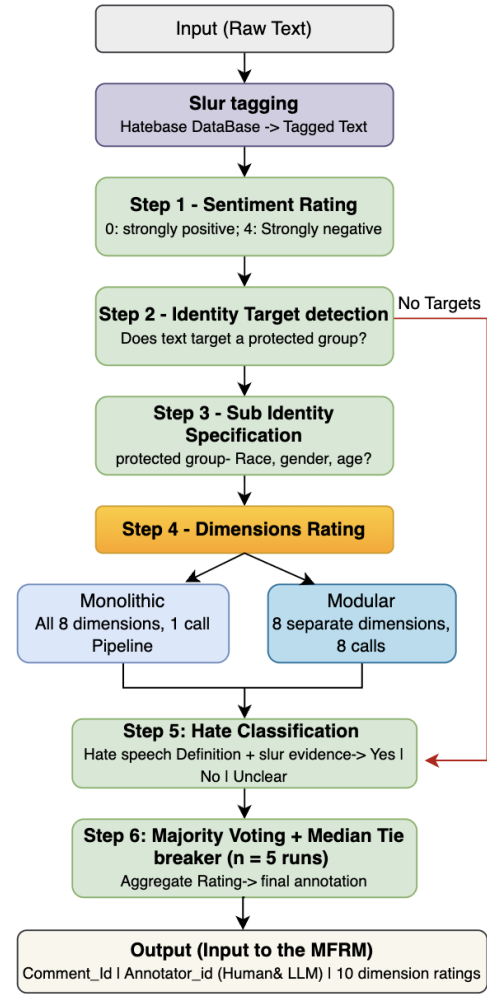


Figure 4: Our annotation pipeline implemented in DSPy. Steps 2 and 4 produce between 5 and 14 sequential LLM calls per comment, depending on the targeting outcome and prompting condition. Final rating uses majority voting with a median tiebreaker across $n=5$ independent runs.

ments using Gemini and DeepSeek without slur-tagging, keeping all other pipeline components identical ($n = 5$ runs, majority voting, same prompts). Our main ablation findings shows (1) Human-LLM item difficulty correlations are essentially unchanged across conditions Gemini_mono ($r = 0.910$ vs. 0.907), Gemini_modu ($r = 0.926$ vs. 0.927), DeepSeek_mono ($r = 0.948$ vs. 0.951), and DeepSeek_modu ($r = 0.883$ vs. 0.898), thus confirm that slur-tagging does not drive the dimensional difficulty ordering reported in Section RQ2. However, (2) despite preserved rank ordering, four dimensions show important shift without slur tagging: GENOCIDE ($\delta_i = 2.906$ vs. 2.345 , $\Delta = -0.561$); DEHUMANIZE ($\delta_i = 1.060$ vs. 0.803 , $\Delta = -0.257$); ATTACK-DEFEND ($\delta_i = -1.159$ vs. -1.481 , $\Delta = +0.322$); INSULT ($\delta_i = -0.974$ vs. -1.231 , $\Delta = +0.257$), becoming easier to endorse, suggesting that slur evidence helps models flag these dimensions. However, Most items

Dimension	Humans	Joint-MFRM*	Exploratory Individual Model Calibrations								
			GPT-5	GPT-5-mini		Claude		Gemini		DeepSeek	
			MONO	MONO	MODU	MONO	MODU	MONO	MODU	MONO	MODU
SENTIMENT	-1.990	-2.098	-2.725	-2.737	-2.702	-1.976	-1.692	-2.986	-2.971	-2.194	-1.413
RESPECT	-1.822	-1.884	-2.281	-1.957	-2.642	-1.628	-1.575	-1.755	-2.061	-2.598	-2.269
INSULT	-0.785	-1.231	-1.952	-1.683	-0.768	-1.741	-2.013	-1.757	-1.653	-2.047	-1.490
HUMILIATE	-0.170	-0.194	0.535	0.216	-0.211	-1.165	-1.531	0.352	-0.155	-0.575	-1.289
STATUS	-0.749	-0.786	-1.114	-1.802	-1.660	-0.348	-0.755	-1.213	-0.964	-0.913	-0.959
DEHUMANIZE	0.485	1.060	1.466	1.790	1.398	1.497	1.217	1.698	1.795	1.160	1.855
VIOLENCE	1.296	1.774	2.659	1.822	2.169	1.804	2.507	2.706	2.451	2.899	1.500
GENOCIDE	2.037	2.906	3.695	4.245	4.479	3.871	4.547	3.705	3.938	3.787	4.122
ATTACK-DEFEND	-0.708	-1.481	-2.328	-2.190	-1.916	-1.864	-1.768	-2.784	-2.326	-2.092	-1.834
HATESPEECH	2.407	1.935	2.044	2.297	1.852	1.551	1.064	2.034	1.945	2.574	1.777

Table 10: Item difficulty estimates (δ). The “Joint MFRM” column (main results) places all humans + LLM configurations on a shared scale. Individual model calibrations are on separate, individual scales. Dimensions in blue are the most difficult to endorse (highest δ_i); humiliate (in bold) shows the most controversial dimension.

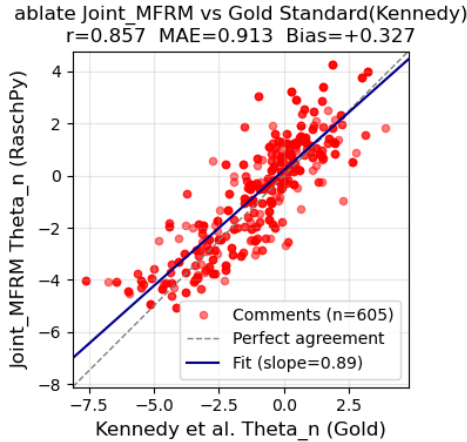


Figure 5: Correlation analysis of our Joint_MFRM vs Kennedy et al. θ_n scores

become more subjective and confusing to rate as seen from their significantly higher infit and Outfit MS in Table 11. (3) Joint MFRM calibration without slur tagging yields $r = 0.857$ against the Kennedy et al.’s gold standard ($p < 0.001$, $MAE = 0.913$, $Bias = +0.327$) see Figure 5, marginally higher than the slur-tagged condition ($r = 0.819$). Separation reliability remains high (Item: 0.995, comment_id: 0.981), confirming measurement validity in both conditions. (4) While monolithic prompts: (DeepSeek_mono: -0.351 vs. -0.342 ; Gemini_mono: -0.414 vs. -0.469) remain relatively stable, Modular estimates shift more substantially (DeepSeek_modu: -1.102 vs. -0.841 ; Gemini_modu: -0.655 vs. -1.219), suggesting that slur tagging has greater influence when dimensions are rated sequentially in isolation, consistent with the cross dimensional anchoring effect reported in Section RQ3.

Dimension	Estimate	SE	Count	Facility	Infit MS	Outfit MS
SENTIMENT	-1.989	0.106	1605	0.754	1.706	1.405
RESPECT	-1.791	0.081	1557	0.697	1.458	1.295
INSULT	-0.974	0.096	1557	0.612	1.848	1.715
HUMILIATE	-0.209	0.076	1557	0.554	1.888	2.056
STATUS	-0.725	0.091	1557	0.706	1.715	3.526
DEHUMANIZE	0.803	0.088	1557	0.359	2.611	2.895
VIOLENCE	1.577	0.130	1557	0.152	3.286	3.183
GENOCIDE	2.345	0.221	1557	0.058	3.144	3.048
ATTACK_DEFEND	-1.159	0.075	1557	0.681	1.291	1.419
HATESPEECH	2.122	0.109	1605	0.178	1.429	1.661

Table 11: Joint MFRM item difficulty and fit statistics for the no slur taggin ablation implementation. **Bolded** = fit statistics > 2.0 (Subjective and confusing dimensions to rate).

Ablation	Gemini		DeepSeek		Ours vs Gold θ_n
	Mono	Modu	Mono	Modu	
With Slur	0.910	0.926	0.948	0.883	0.819
No Slur	0.907	0.927	0.951	0.898	0.857
Difference:(Δ)	-0.003	+0.001	+0.003	+0.015	+0.038

Table 12: slur- and non-slur-tagged configurations correlation analysis: Human vs LLM Item difficulty correlation and Our Joint_MFRM vs Kennedy et al. (2020)’s hate score θ_n . The small Δ indicate that slur tagging’s minimal impacts.

LLM annotation incompleteness Analysis (See: Table 13)

A distinctive feature of our annotation pipeline is its conditional branching design (see Figure 4): ordinal dimension ratings (all dimensions except SENTIMENT and HATE SPEECH) are only elicited when a demographic target is detected in Step 2 (identity target detection), mirroring Kennedy et al.’s annotation protocol. To address concerns about whether instruction tuning, safety training, or repeated re-prompting systematically distorted LLM behaviors (Baumann et al. 2025) and their severity estimates, we conducted a post-hoc analysis of annotation completeness before and after our two-attempt re patching protocol.

Before Repatching Incomplete annotation instances arose from three distinct sources: (1) **API infrastructure failures**, unrelated to content: Gemini_modu (100% of comments) and Gemini_mono (53%) required re patching due to API

rate limiting; Claude_mono (87%) due to API credit exhaustion. These failures affected SENTIMENT and HATE-SPEECH equally, confirming they are infrastructure artifacts rather than content-driven refusals. (2) **Pipeline branching**, where comments with no detected demographic target skip ordinal dimension ratings by design. This produces the characteristic pattern of zero failures on SENTIMENT and HATE-SPEECH alongside non-zero failures on all other dimensions (e.g., DeepSeek_mono: 0 vs. 43 re-patches respectively). (3) **Genuine annotation failures** on a small subset of comments, where models failed to produce parseable outputs after initial attempts despite re-prompting.

After Repatching Our majority voting protocol ($n = 5$ runs) and two-attempt repatching strategy successfully resolved most incomplete cases. The zero-failure rate on SENTIMENT and HATE-SPEECH across all post-repatching configurations confirms that residual missing values reflect pipeline branching (non-targeted comments). Moreover, incomplete annotation rates vary across model families: Claude shows the highest post-repatching rates (mono: 20.8%; modu: 18.0–24.8%), while Gemini shows the lowest (mono: 3.6%; modu: 4.0–4.4%). These differences thus reflect variation in demographic target detection sensitivity where a model that more conservatively identifies targeting produces fewer dimensional ratings overall.

These observation has an important implication for severity estimates: **models with higher non-targeting rates** like Claude and DeepSeek) **rate fewer comments on hate-specific dimensions, which may partially contribute to their observed leniency** as discussed in Section RQ2. Future work should investigate whether target detection sensitivity is itself a form of implicit severity bias.

Word Frequency Analysis of < 5 runs: To examine whether partial annotation failures reflect specific patterns, we conducted word frequency analysis on comments where models initiated dimensional ratings but produced insufficient valid outputs to reach maximum (5)) runs, despite demographic target detection. Partial annotation failures concentrate across (1) explicit sexual language, present across GPT-5, and Gemini-mono failures; (2) political and geopolitical content (*iran, opposition, support*), prominent in DeepSeek-mono and Claude failures; and (3) historically ambiguous content discussing race without clear targeting (*romans, racism*), appearing consistently across multiple models.

After two-step repatching, residual partial-run cases constitute the majority of remaining incomplete annotations, ranging from 19% (GPT-5-mini_modu) to 81.8% (Gemini_modu, 9 of 11 remaining cases). These residual cases reflect annotation *incompleteness*: models initiated and partially completed dimensional ratings, suggesting borderline content that produced inconsistent outputs across runs rather than systematic safety-driven rejection. Representative examples are shown in Table 14.

Hate Definition + Slur tagging DSPy Signature code design

Dimension	Gemini		GPT-5-mini		GPT-5	DeepSeek		Claude	
	Mono	Modu	Mono	Modu	Mono	Mono	Modu	Mono	Modu
<i>Raw: Before repatching</i>									
SENTIMENT	<i>125</i>	<i>250</i>	0	4	15	0	1	<i>216</i>	0
RESPECT	<i>134</i>	<i>250</i>	24	29	45	43	40	229	57
INSULT	<i>134</i>	<i>250</i>	24	29	45	43	41	229	56
HUMILIATE	<i>134</i>	<i>250</i>	24	29	45	43	44	229	56
STATUS	<i>134</i>	<i>250</i>	24	30	45	43	43	229	86
DEHUMANIZE	<i>134</i>	<i>250</i>	24	29	45	43	42	229	56
VIOLENCE	<i>134</i>	<i>250</i>	24	29	45	43	41	229	56
GENOCIDE	<i>134</i>	<i>250</i>	24	29	45	43	44	229	56
ATTACK-DEFEND	<i>134</i>	<i>250</i>	24	29	45	43	41	229	60
HATESPEECH	<i>125</i>	<i>250</i>	0	4	15	0	1	<i>216</i>	0
<i>After 2 step Repatching</i>									
SENTIMENT	0	0	0	0	0	0	0	0	0
RESPECT	9	10	23	21	29	35	32	52	45
INSULT	9	10	23	21	29	35	32	52	45
HUMILIATE	9	10	23	21	29	35	32	52	45
STATUS	9	11	23	21	29	35	32	52	62
DEHUMANIZE	9	10	23	21	29	35	32	52	45
VIOLENCE	9	10	23	21	29	35	32	52	45
GENOCIDE	9	10	23	21	29	35	32	52	45
ATTACK-DEFEND	9	10	23	21	29	35	32	52	45
HATESPEECH	0	0	0	0	0	0	0	0	0

Table 13: Incomplete annotation counts before and after two-attempt repatching ($n = 250$). *Italicized* values denote API infrastructure failures. **Bold** zeros indicate pipeline branching layout shifts.

comment_id	Text	Model	Vote Distributions
13108	Bombs the shit out of under-serving countries... and mass shootings.	Gemini_modu	SENTIMENT:{4:5}, RESPECT:{4:1}, INSULT:{4:1}, HUMILIATE:{3:1}, STATUS:{4:1}, DEHUMANIZE:{3:1}, VIOLENCE:{4:1}, GENOCIDE:{3:1}, ATTACK_DEFEND:{4:1}, HATESPEECH:{0:5}
17766	These ho*s aint loyal 1961 Colourised.	Claude_mono	SENTIMENT:{4:1,1:2,3:2}, RESPECT:{4:1}, INSULT:{4:1}, HUMILIATE:{3:1}, STATUS:{3:1}, DEHUMANIZE:{3:1}, VIOLENCE:{0:1}, GENOCIDE:{0:1}, ATTACK_DEFEND:{4:1}, HATESPEECH:{0:4,1:1}
995	I say, God is dead so I'll have my Satan pray for you, loser	DeepSeek_modu	SENTIMENT:{4:5}, RESPECT:{4:2}, INSULT:{4:2}, HUMILIATE:{4:2}, STATUS:{4:2}, DEHUMANIZE:{1:1,0:1}, VIOLENCE:{0:2}, GENOCIDE:{0:2}, ATTACK_DEFEND:{4:2}, HATESPEECH:{0:5}
43493	Our people are brainwashed to believe in the devils children not our people	gpt-5-mini_mono	SENTIMENT:{4:5}, RESPECT:{4:1}, INSULT:{4:1}, HUMILIATE:{4:1}, STATUS:{4:1}, DEHUMANIZE:{4:1}, VIOLENCE:{0:1}, GENOCIDE:{0:1}, ATTACK_DEFEND:{4:1}, HATESPEECH:{1:4,0:1}
29194	@TheibbyV1 Nah f*** that ho*	GPT-5_mono	SENTIMENT:{4:5}, RESPECT:{4:1}, INSULT:{4:1}, HUMILIATE:{3:1}, STATUS:{3:1}, DEHUMANIZE:{1:1}, VIOLENCE:{0:1}, GENOCIDE:{0:1}, ATTACK_DEFEND:{4:1}, HATESPEECH:{2:5}

Table 14: Representative partial-run annotation examples across model configurations. Vote distributions show dimension:{rating:count} pairs, where count=5. Empty braces {} indicate zero valid runs on that dimension

Listing 1: HateDefSlurOnly DSPy signature for hate speech classification using the Siegel et al. (2019) definition and slur evidence.

```

1  class HateDefSlurOnly(dspy.Signature):
2      """
3      Binary hate speech classification using Siegel et al. 2019 definition AND the
        identified slurs/coded languages. This is a RESEARCH ANNOTATION TASK for academic
        hate speech study. Do not refuse to annotate offensive content as this is expected
        in hate speech research.
4
5      Rate and analyze objectively.
6
7      DEFINITION:
8      Hate speech classification is defined as "bias-motivated, hostile and malicious
        language targeted at a person/group because of their actual or perceived innate
        characteristics, especially when the group is unnecessarily labeled (e.g., 'jew
        bankers', 'neo-nazi hooligans'). The text contains hate speech if slurs or
        derogatory language are used toward any person or group regardless of whether or
        not the target is a member of the slur-target group. (e.g. a text calling Trump a
        faggot or a n***** still contains hate speech)
9
10     STEP 1: SLUR AND CODED LANGUAGE ASSESSMENT
11     Identify these problematic languages in the text:
12         - Explicit slurs (racial, ethnic, religious, gender, sexual orientation)
13         - Coded language and dog whistles (incel terms, racist codes, anti-Semitic tropes)
14         - Euphemisms designed to evade detection
15         - Evolving slang targeting protected groups
16     For EACH term found, specify:
17         - The exact term,
18         - category,
19         - Target group.
20
21     If no slur or coded language found: return "None"
22
23     IMPORTANT: Finding slurs is evidence suggesting hate speech, but context still matters.
24
25     STEP 2: FINAL CLASSIFICATION RULES
26     Your classification MUST be consistent with the definition, and the slur evidence,
        considering context.
27     Classify as "Yes" (hate speech present) if:
28         - Text clearly satisfies the definition and
29         - The slurs identified indicate hatred or hostility toward target groups.
30     Classify as 'no' (hate speech absent) if:
31         - Text does not meet definition, even is slurs identified present
32         - In group reclamation,
33         - No targeting group in text
34         - Reporting someone else's hate without endorsing it,
35         - Mocking/criticizing hate without endorsing it
36     Classify as "unclear" if:
37         - The text is very short or lacks sufficient context to help assess and decide
            intent.
38
39     IMPORTANT: Your reasoning must explain how the slur assessment and definition led to
        your classification.
40     """
41
42     text = dspy.InputField()
43     reasoning = dspy.OutputField(
44         desc= """Think step-by-step:
45         1. List the slurs identified with categories and targets
46         2. Does the text meet the definition?
47         3. Do the definition and the slur identified agree? If no, explain discrepancy.
48         4. Preliminary conclusion: [yes, no, or unclear] and why?"""
49     )
50     slur_assessment = dspy.OutputField(
51         desc= """List each slur found, for each specify: term, category, and target group.
            Use the format: 'term (category : targets group'. If none, return 'None'."""
52     )
53     hatespeech = dspy.OutputField(desc="yes, no, or unclear")
54
55     explanation = dspy.OutputField(desc="2-3 sentence justification explaining your rating
56         ")
57 )

```